EMPLOYEE ATTRITION ANALYSIS

Insights & Recommendations for Employee Retention Prediction

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BUSINESS CONTEXT

Why Employee Attrition Matters

Losing employees is one of the costliest, least visible drains on a growing organization. Before diving into the data, here is the financial backdrop that makes this analysis worth acting on.



15–20%

of an employee's annual salary is spent recruiting their replacement



52 days

average time it takes to fill one open position



1–2.5%

of total company revenue lost to ramp-up time for new hires



40%

of a small-business owner's working hours spent on non-revenue tasks like hiring

THE DATASET

What I Analyzed

1,470

employee records

35

raw data attributes

3

departments covered

Model-ready feature matrix: 45 columns after one-hot encoding

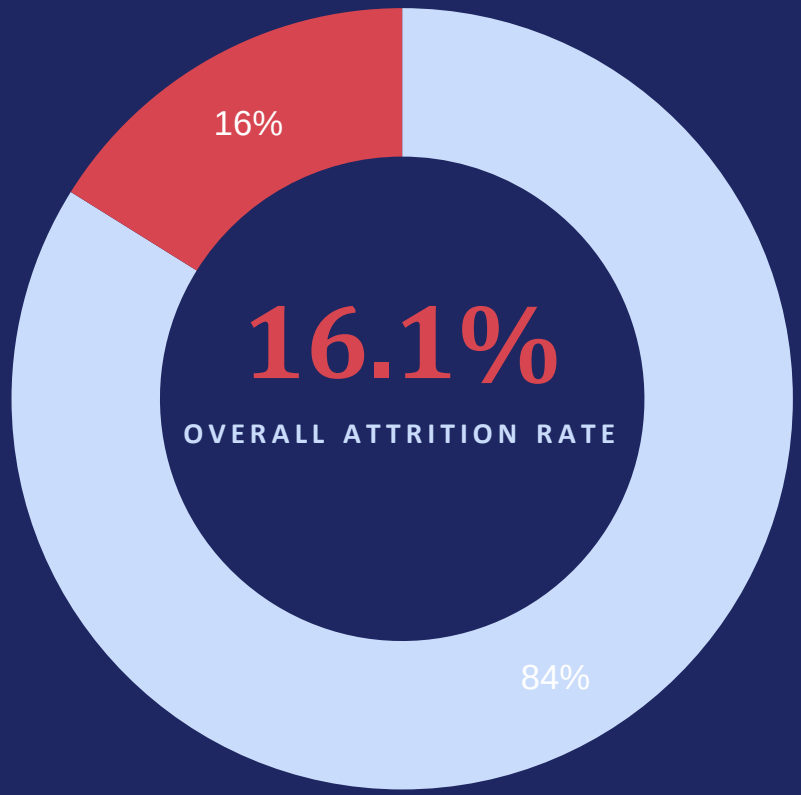
Attribute Categories

- **Demographics** — Age, Gender, Marital Status, Distance from Home
- **Compensation** — Monthly Income, Daily/Hourly Rate, Stock Options, Salary Hike %
- **Job Profile** — Department, Job Role, Job Level, Business Travel, OverTime
- **Engagement** — Job Satisfaction, Involvement, Environment & Relationship Satisfaction, Work-Life Balance
- **Career History** — Years at Company, Years in Role, Since Last Promotion, Companies Worked

Target variable: Attrition (Yes / No) — whether the employee has left the company

HEADLINE FINDING

1 in 6 Employees Left the Company



1,233

employees stayed (83.9%)

237

employees left the company (16.1%)

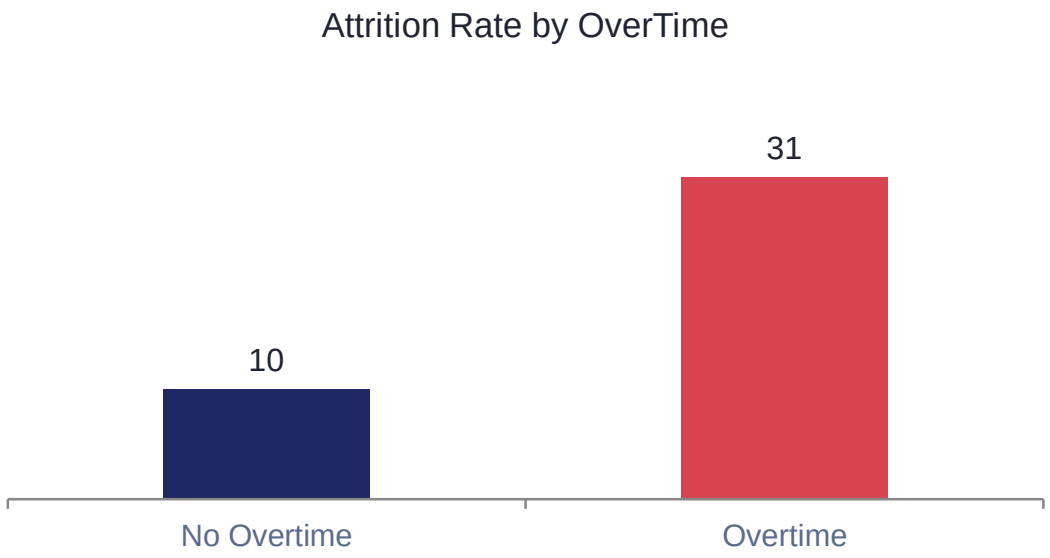
Roughly 1 in every 6

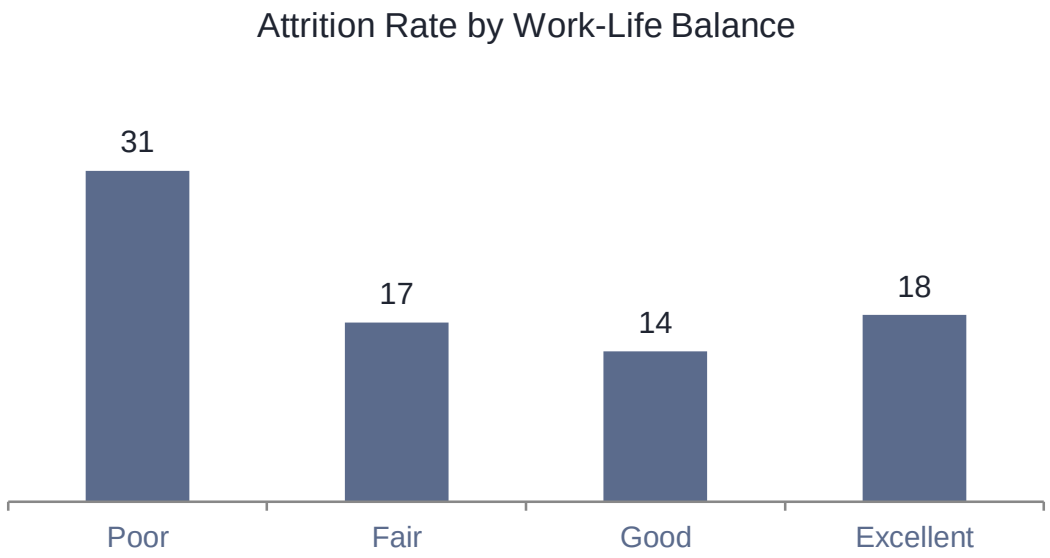
employees in this workforce is an attrition risk today

The sections ahead break down exactly which employee segments drive this number — and why.

DRIVER 01 — WORKLOAD

Overtime Is the #1 Red Flag

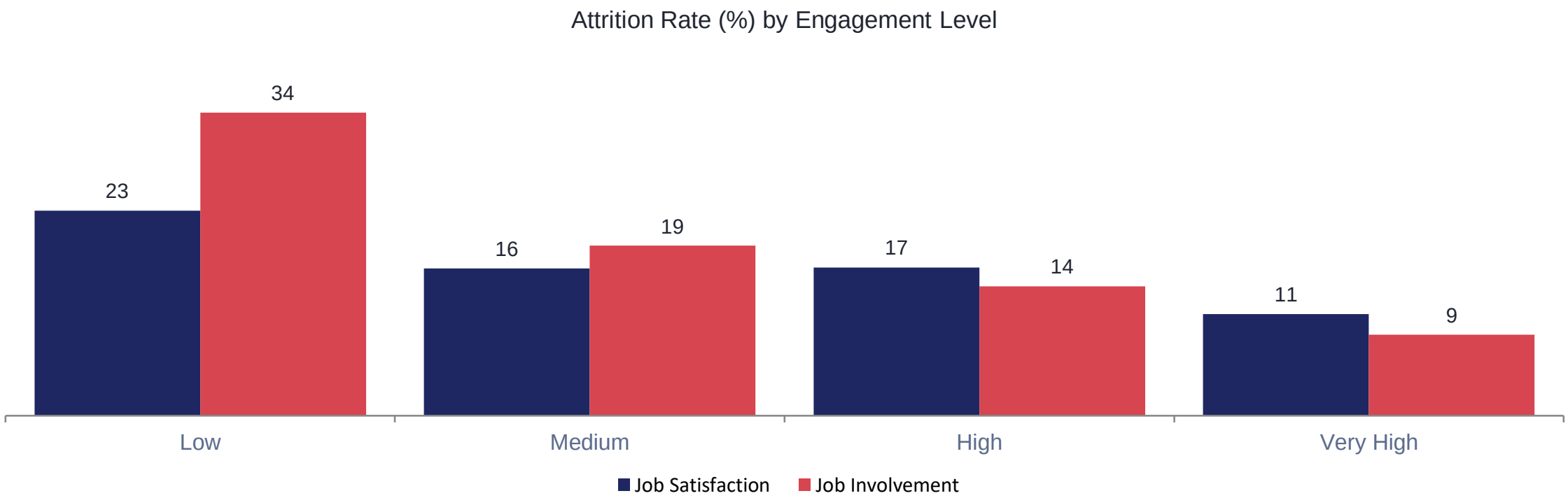




Employees who work overtime leave nearly 3x more often (30.5% vs. 10.4%). Work-life balance shows a similar pattern — the "Poor" rating group attrits at more than double the rate of "Good." Sustained overload is the single strongest behavioral signal of flight risk in this dataset.

DRIVER 02 — ENGAGEMENT

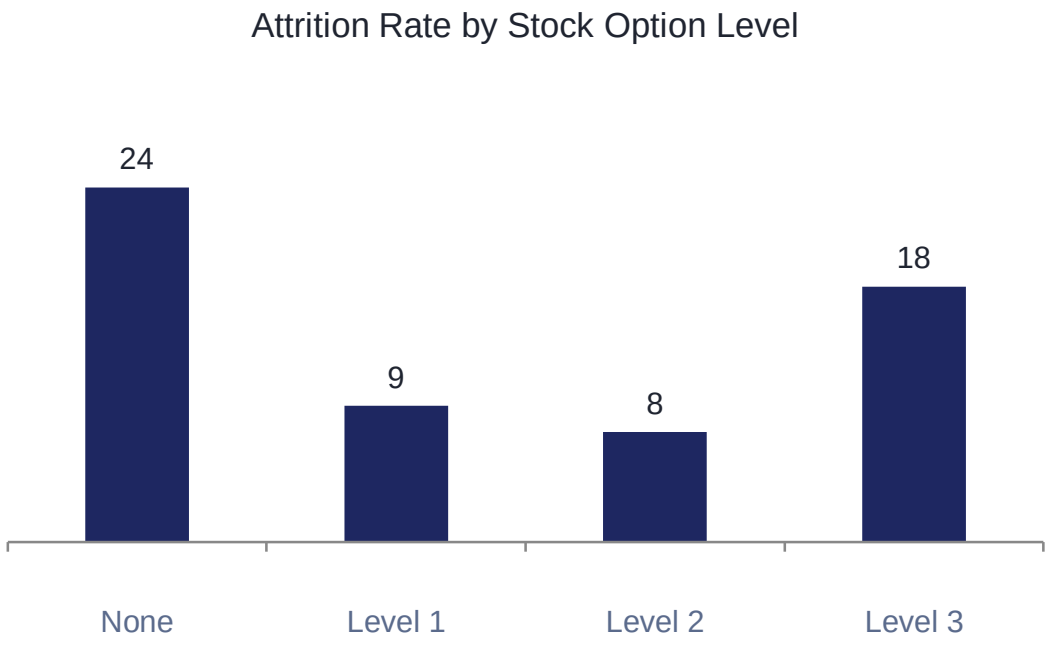
Disengaged Employees Leave First



Both metrics show the same steady decline: attrition among the least-engaged employees (33.7% for low involvement, 22.8% for low satisfaction) is roughly 3x that of the most engaged. Engagement is a leading indicator HR can track continuously.

DRIVER 03 — REWARDS & GROWTH

Pay and Equity Gaps Widen the Risk



Average Monthly Income

$6,833

Employees Who Stayed

$4,787

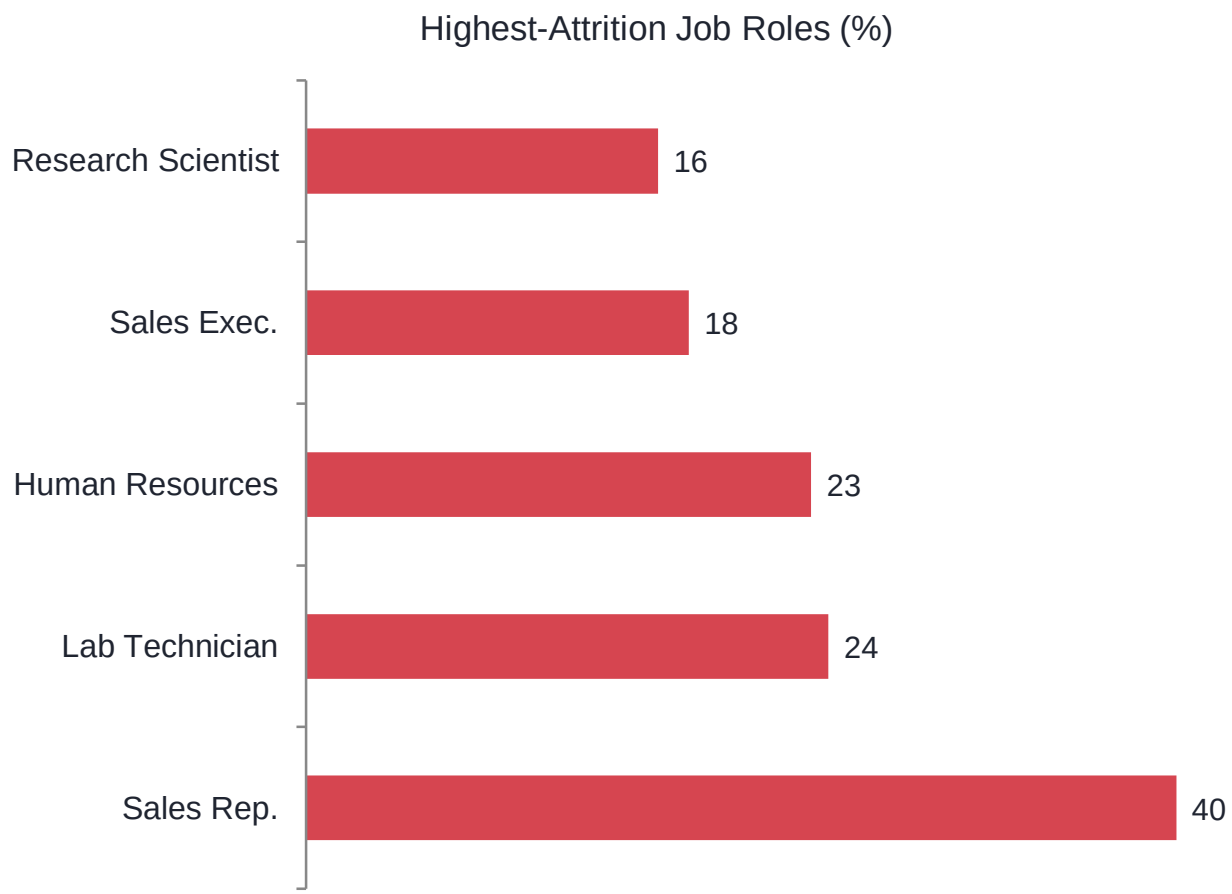
Employees Who Left

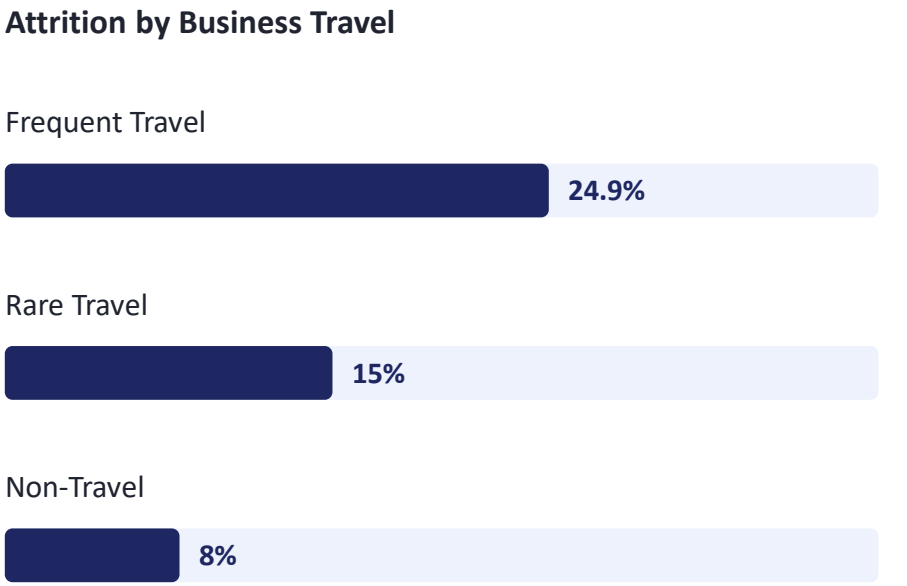
Employees who left earned about 30% less on average than those who stayed.

Employees with zero stock options leave 3x more than those with Level 1–2 grants, and pay is a consistent gap. Equity and compensation review points are natural retention levers.

DRIVER 04 — ROLE & TRAVEL

Some Roles Are Simply Higher Risk

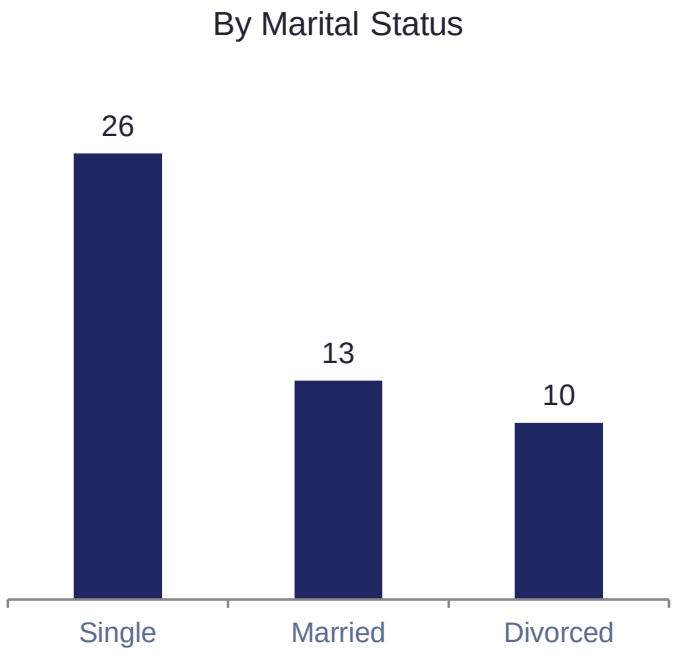




Sales Representatives churn at 39.8% — more than double the company average — and frequent travelers leave 3x more than non-travelers.

DRIVER 05 — LIFE STAGE & TENURE

Younger, Newer, Single Employees Are Flight Risks



Average Age

37.6 yrs	33.6 yrs
Stayed	Left

Average Tenure at Company

7.4 yrs	5.1 yrs
Stayed	Left

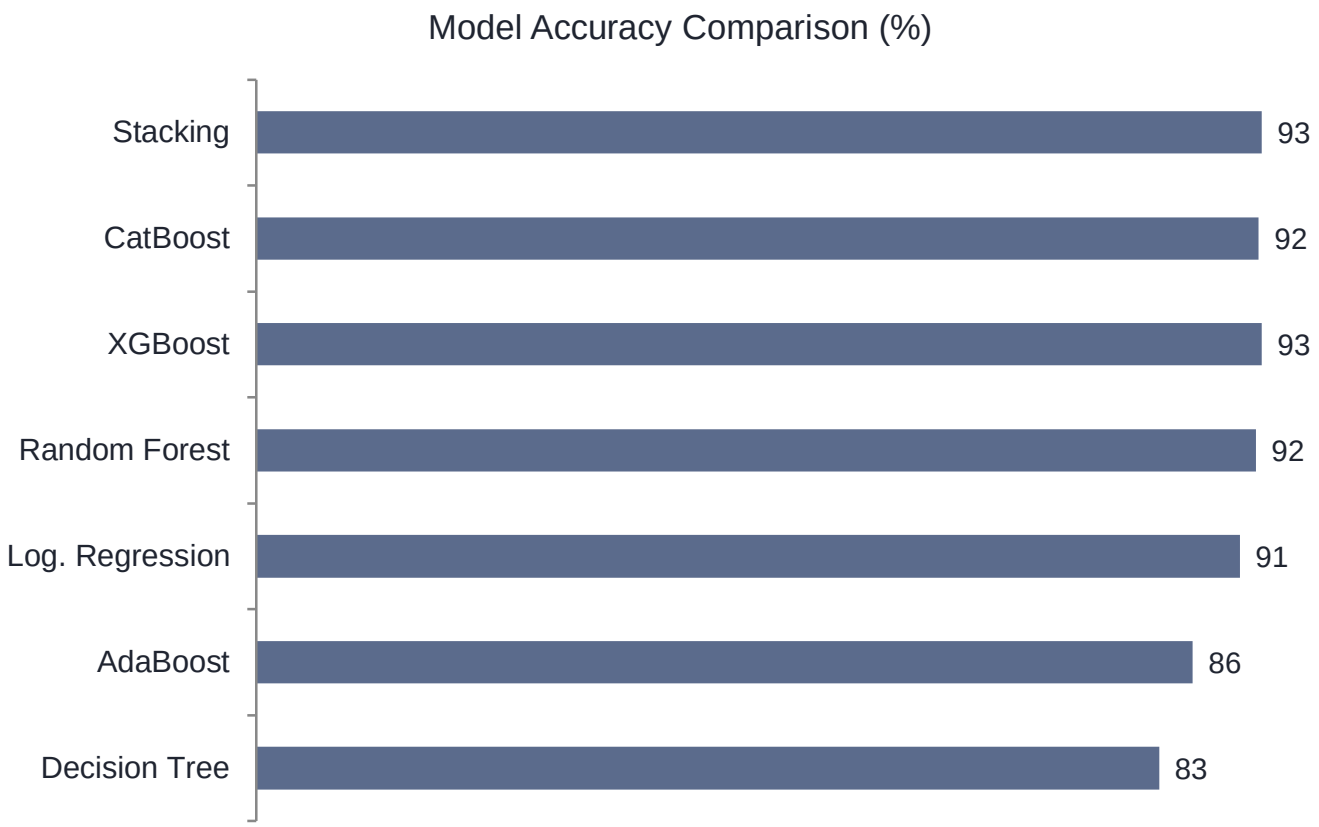
Reading the Pattern

Single employees leave 2–2.5x more often than married or divorced peers. Leavers are, on average, 4 years younger and have 2.3 fewer years of tenure — attrition is concentrated early in the employee lifecycle.

PREDICTIVE MODELING

Can We Predict Who Leaves Next?

Seven classification models were trained on the (SMOTE-balanced) employee data to flag at-risk employees before they resign.



Best Performer

XGBoost

Accuracy	92.6%
Precision	92.7%
Recall	92.6%
F1-Score	92.6%

XGBoost is selected by the notebook's programmatic ranking; it matches Stacking on accuracy and recall (92.6%) and has slightly higher precision.

RECOMMENDATIONS

Turning Insight Into Action



Cap & Monitor Overtime

Set overtime thresholds and review workload for employees consistently logging extra hours — the single largest attrition signal.



Track Engagement Continuously

Run regular satisfaction & involvement pulse surveys; route low scorers to managers for 1:1 check-ins before they disengage further.



Review Pay & Equity Early

Extend stock option eligibility and benchmark pay for early-tenure staff, where the income and equity gap is widest.



Target High-Risk Roles

Build tailored retention plans for Sales Representatives and Lab Technicians, and cap frequent travel where possible.



Support Early-Career Employees

Strengthen onboarding and mentorship in years 1–3, when single and younger employees are most likely to leave.



Deploy the Predictive Model

Score active employees monthly with the XGBoost model (92.6% accuracy) to flag at-risk individuals for proactive HR outreach.

Thank You